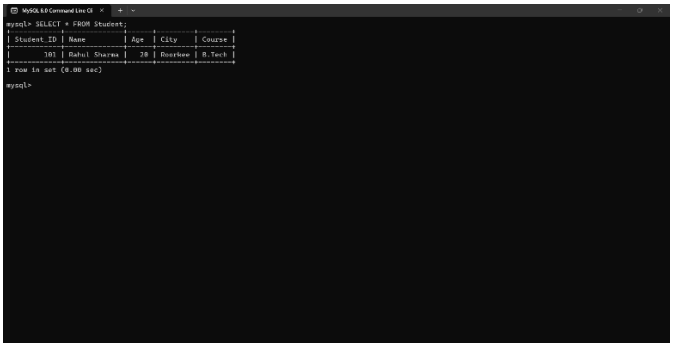
Lab Sheet 2

Q1- Insert one record into the Student table.

Sol1- 

Q2- Insert five student records using separate INSERT statements.

Sol2-

MySQL 8.0 Command Line Cli

```
mysql> SELECT * FROM Student;
+------------+--------------+------+----------+--------+
| Student_ID | Name         | Age  | City     | Course |
+------------+--------------+------+----------+--------+
|        101 | Rahul Sharma |   20 | Roorkee  | B.Tech |
|        102 | Priya Verma  |   19 | Meerut   | BCA    |
|        103 | Amit Kumar   |   21 | Dehradun | B.Tech |
|        104 | Neha Gupta   |   20 | Haridwar | BBA    |
|        105 | Rohit Singh  |   22 | Meerut   | MCA    |
|        106 | Sneha Rawat  |   18 | Dehradun | BCA    |
+------------+--------------+------+----------+--------+
6 rows in set (0.00 sec)

mysql>
```

Q3- Insert multiple records into the Student table using a single statement.

Sol3-

```
MySQL 8.0 Command Line Cli

mysql> INSERT INTO Student VALUES
    ->     (107, 'Vikas Yadav',  20, 'Haridwar', 'B.Tech'),
    ->     (108, 'Pooja Negi',   19, 'Roorkee',  'BBA'),
    ->     (109, 'Ankit Tyagi',  21, 'Meerut',   'B.Tech'),
    ->     (110, 'Kavya Joshi',  18, 'Dehradun', 'MCA');
Query OK, 4 rows affected (0.00 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql> SELECT * FROM Student;
+------------+---------------+------+----------+---------+
| Student_ID | Name          | Age  | City     | Course |
+------------+---------------+------+----------+---------+
|        101 | Rahul Sharma  |   20 | Roorkee  | B.Tech |
|        102 | Priya Verma   |   19 | Meerut   | BCA    |
|        103 | Amit Kumar    |   21 | Dehradun | B.Tech |
|        104 | Neha Gupta    |   20 | Haridwar | BBA    |
|        105 | Rohit Singh   |   22 | Meerut   | MCA    |
|        106 | Sneha Rawat   |   18 | Dehradun | BCA    |
|        107 | Vikas Yadav   |   20 | Haridwar | B.Tech |
|        108 | Pooja Negi    |   19 | Roorkee  | BBA    |
|        109 | Ankit Tyagi   |   21 | Meerut   | B.Tech |
|        110 | Kavya Joshi   |   18 | Dehradun | MCA    |
+------------+---------------+------+----------+---------+
10 rows in set (0.00 sec)

mysql>
```

Q4- Insert records into the Employee table.

Sol4-

```
MySQL 8.0 Command Line Cli

mysql> CREATE TABLE Employee (
    ->     Emp_ID     INT PRIMARY KEY,
    ->     Name       VARCHAR(50),
    ->     Department VARCHAR(30),
    ->     Salary     DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.03 sec)

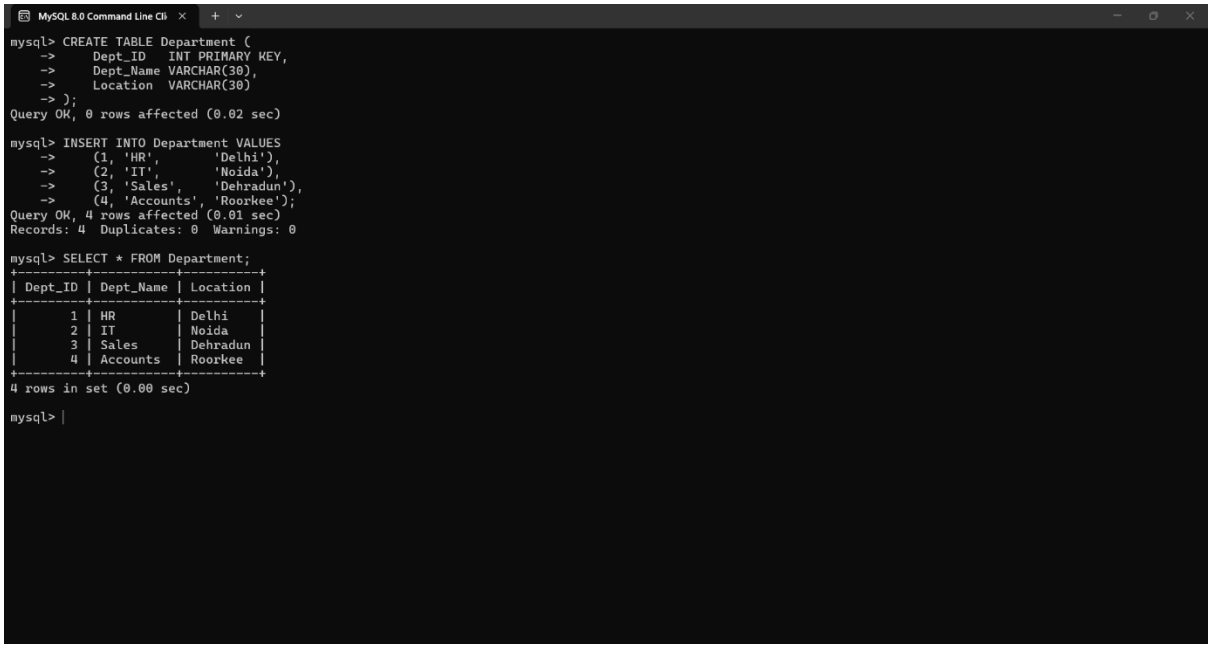
mysql> INSERT INTO Employee VALUES
    ->     (201, 'Suresh Kumar',  'HR',         28000),
    ->     (202, 'Anita Sharma',  'IT',         45000),
    ->     (203, 'Manoj Verma',   'Sales',      25000),
    ->     (204, 'Ritu Singh',    'IT',         60000),
    ->     (205, 'Deepak Chauhan','Accounts',   35000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql> SELECT * FROM Employee;
+--------+---------------+------------+----------+
| Emp_ID | Name          | Department | Salary   |
+--------+---------------+------------+----------+
|    201 | Suresh Kumar  | HR         | 28000.00 |
|    202 | Anita Sharma  | IT         | 45000.00 |
|    203 | Manoj Verma   | Sales      | 25000.00 |
|    204 | Ritu Singh    | IT         | 60000.00 |
|    205 | Deepak Chauhan| Accounts   | 35000.00 |
+--------+---------------+------------+----------+
5 rows in set (0.01 sec)

mysql>
```

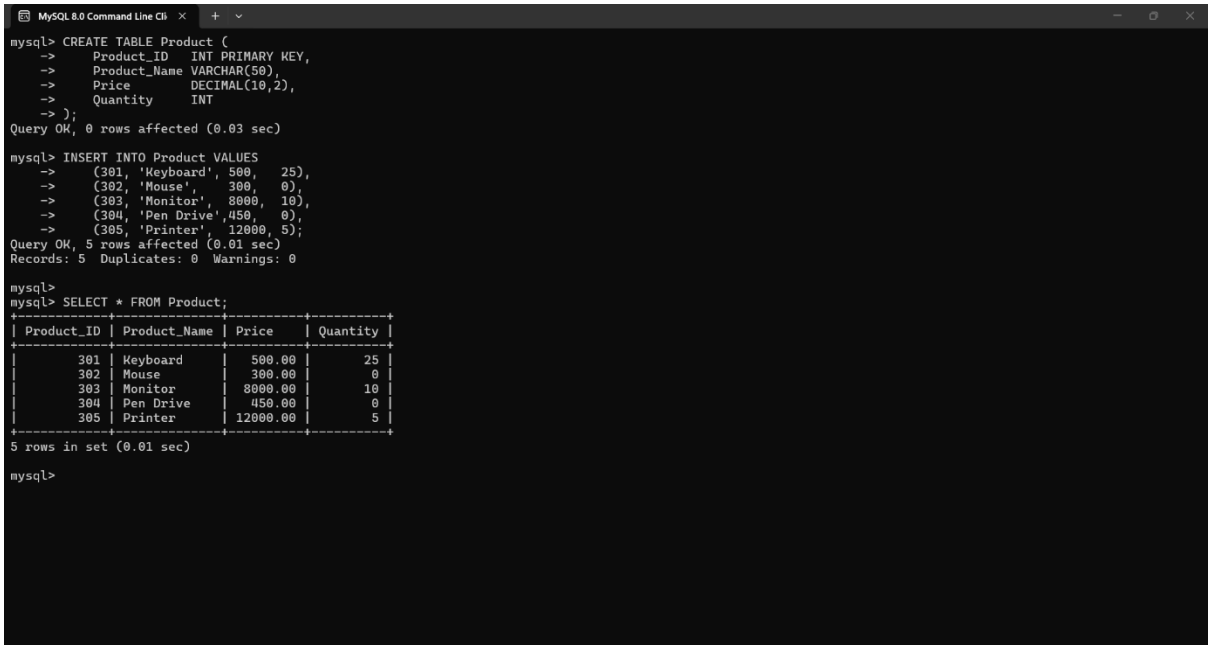
Q5- Insert records into the Department table

Sol5-



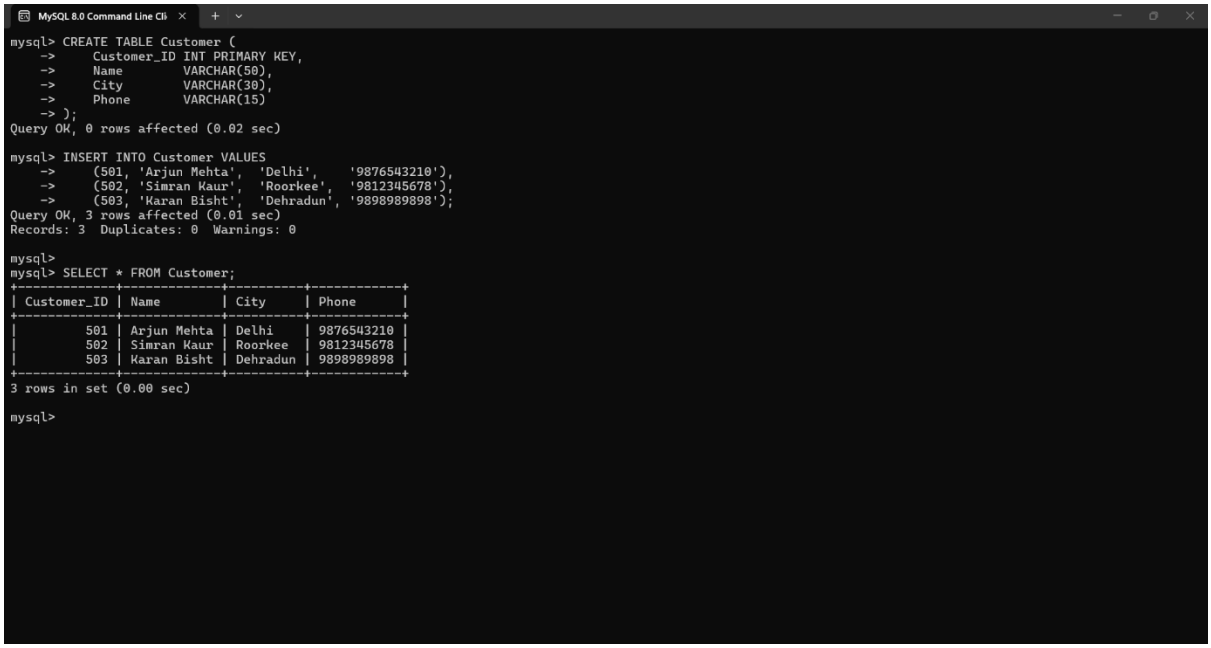
Q6- Insert records into the Product table.

Sol6-



Q7- Insert records into the Customer table.

Sol7-



```
MySQL 8.0 Command Line Cli

mysql> CREATE TABLE Customer (
    ->     Customer_ID INT PRIMARY KEY,
    ->     Name        VARCHAR(50),
    ->     City        VARCHAR(30),
    ->     Phone       VARCHAR(15)
    -> );
Query OK, 0 rows affected (0.02 sec)

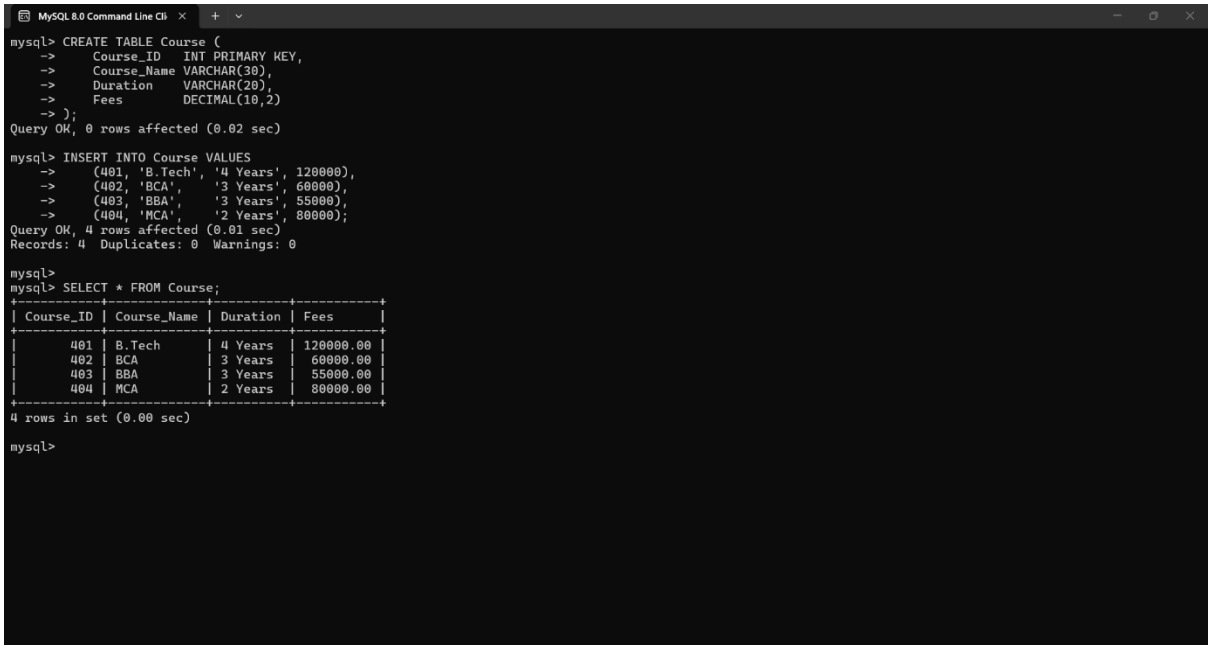
mysql> INSERT INTO Customer VALUES
    ->     (501, 'Arjun Mehta',  'Delhi',    '9876543210'),
    ->     (502, 'Simran Kaur',  'Roorkee',  '9812345678'),
    ->     (503, 'Karan Bisht',  'Dehradun', '9898989898');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> SELECT * FROM Customer;
+-------------+-------------+----------+------------+
| Customer_ID | Name        | City     | Phone      |
+-------------+-------------+----------+------------+
|         501 | Arjun Mehta | Delhi    | 9876543210 |
|         502 | Simran Kaur | Roorkee  | 9812345678 |
|         503 | Karan Bisht | Dehradun | 9898989898 |
+-------------+-------------+----------+------------+
3 rows in set (0.00 sec)

mysql>
```

Q8- Insert records into the Course table.

Sol8-



```
MySQL 8.0 Command Line Cli

mysql> CREATE TABLE Course (
    ->     Course_ID   INT PRIMARY KEY,
    ->     Course_Name VARCHAR(30),
    ->     Duration    VARCHAR(20),
    ->     Fees        DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.02 sec)

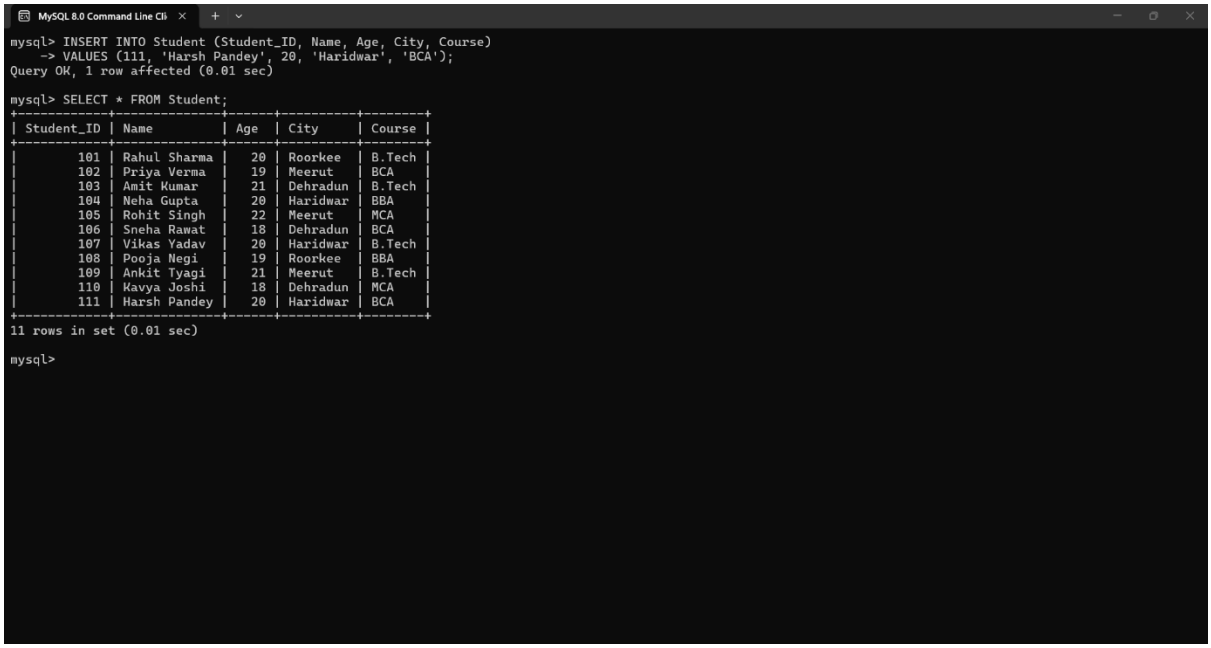
mysql> INSERT INTO Course VALUES
    ->     (401, 'B.Tech', '4 Years', 120000),
    ->     (402, 'BCA',    '3 Years', 60000),
    ->     (403, 'BBA',    '3 Years', 55000),
    ->     (404, 'MCA',    '2 Years', 80000);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql> SELECT * FROM Course;
+-----------+-------------+----------+-----------+
| Course_ID | Course_Name | Duration | Fees      |
+-----------+-------------+----------+-----------+
|       401 | B.Tech      | 4 Years  | 120000.00 |
|       402 | BCA         | 3 Years  |  60000.00 |
|       403 | BBA         | 3 Years  |  55000.00 |
|       404 | MCA         | 2 Years  |  80000.00 |
+-----------+-------------+----------+-----------+
4 rows in set (0.00 sec)

mysql>
```

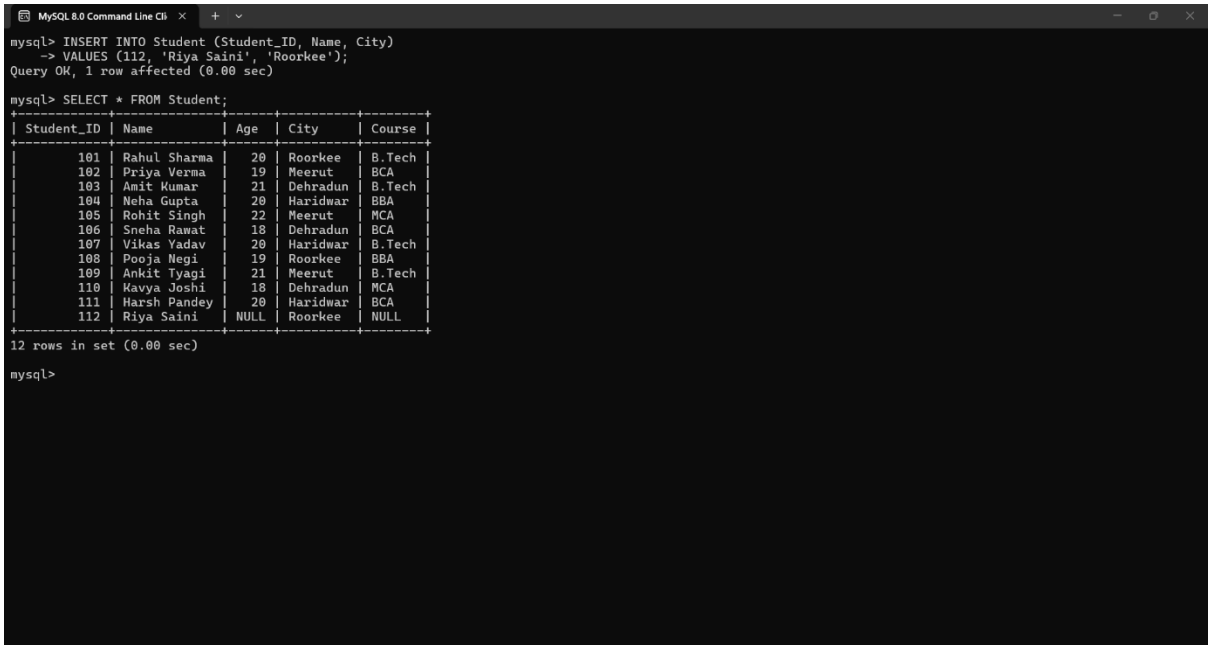
Q9- Insert a student record with values for all columns.

Sol9-



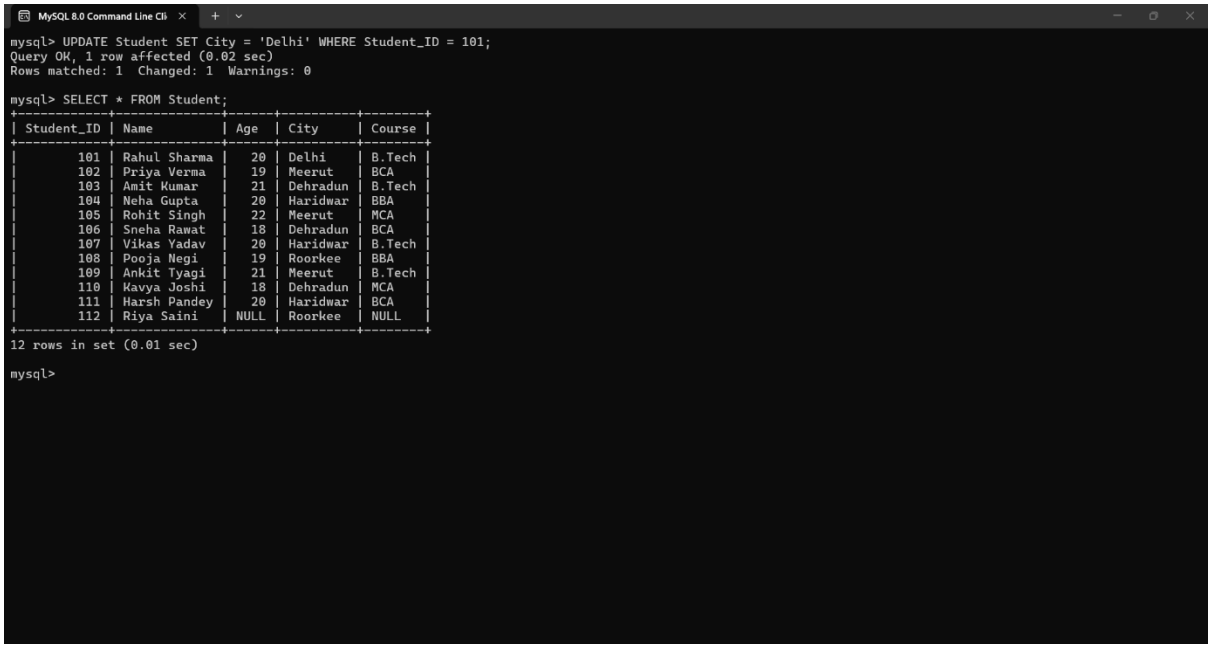
Q10- Insert a student record by specifying only selected columns.

Sol10-



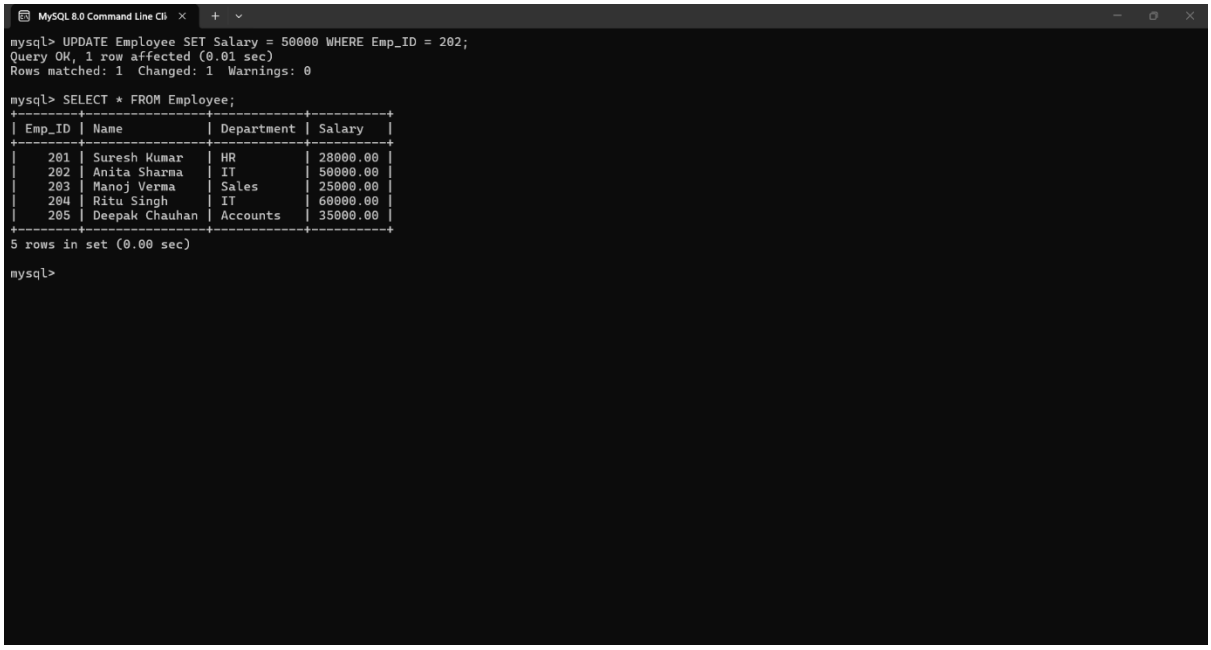
Q11- Update the city of a particular student.

Sol11-



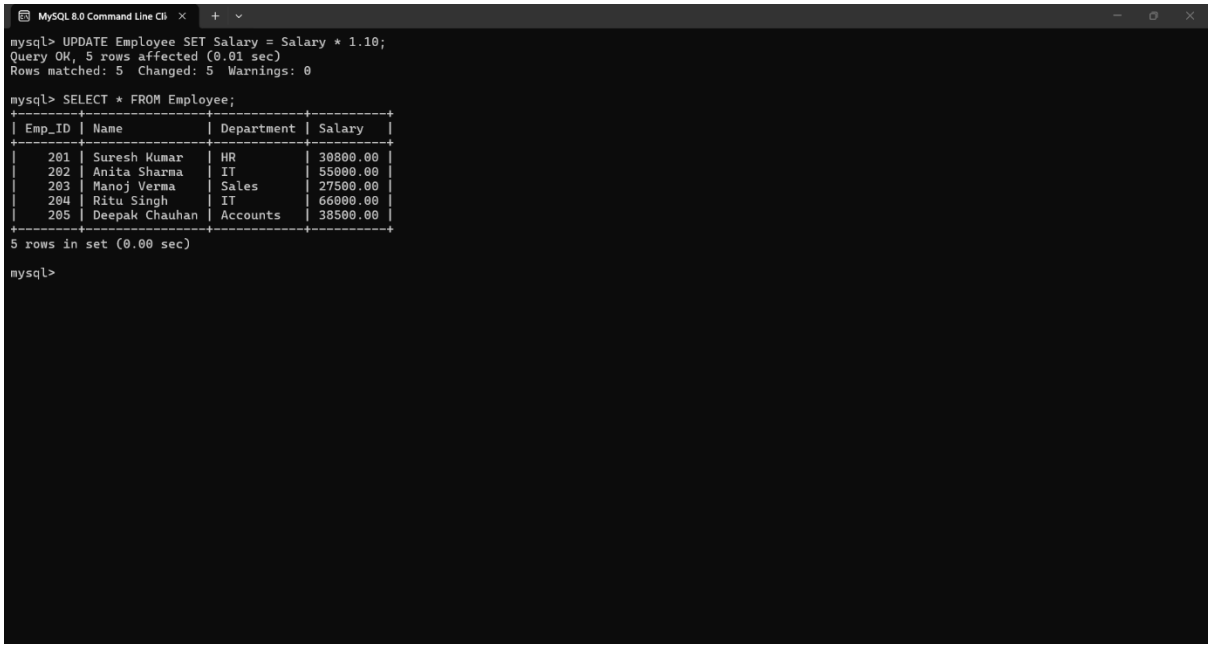
Q12- Update the salary of a particular employee.

Sol12-



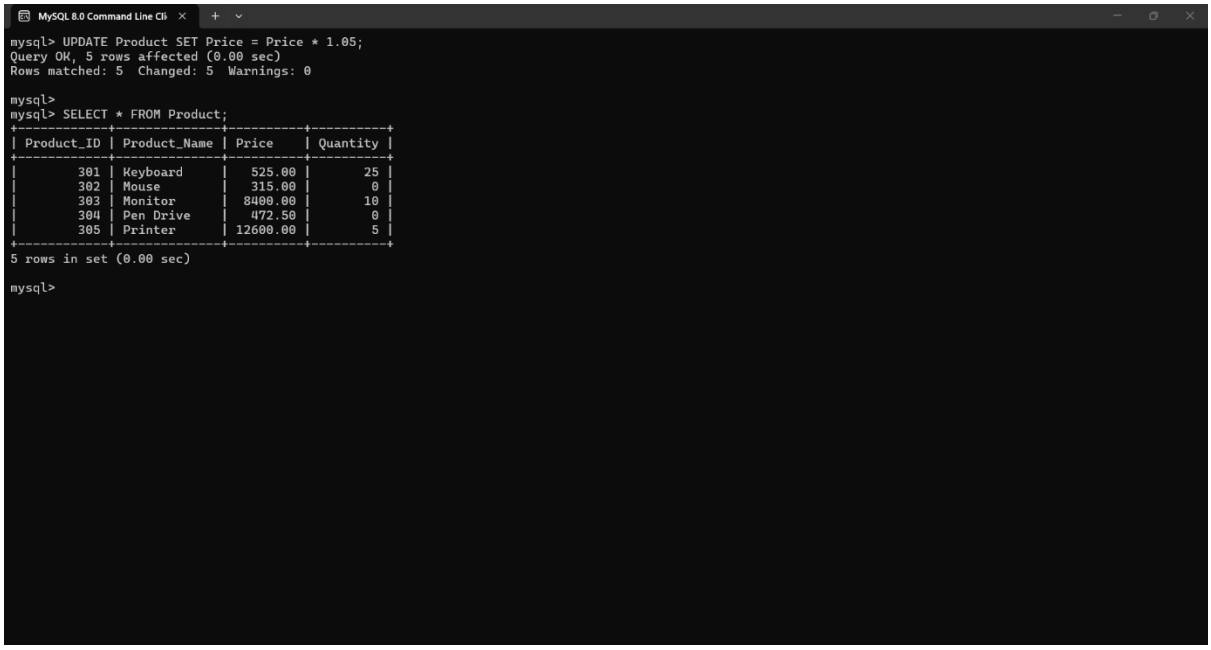
Q13- Increase the salary of all employees by 10%.

Sol13-



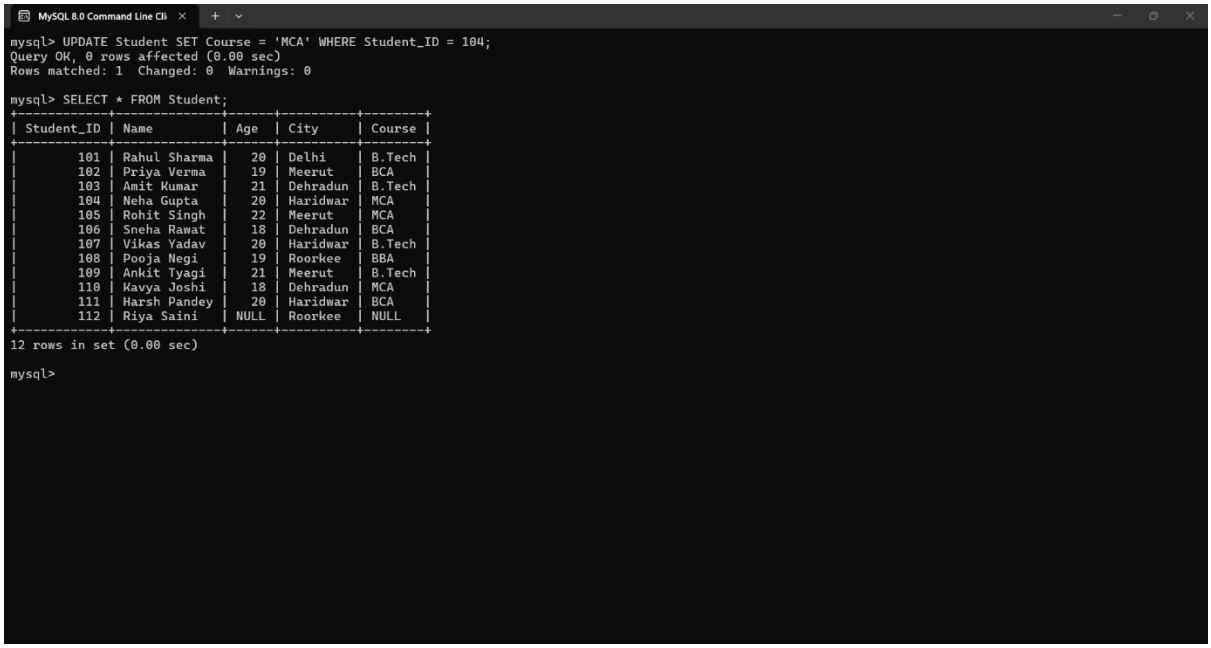
Q14- Increase the price of all products by 5%.

Sol14-



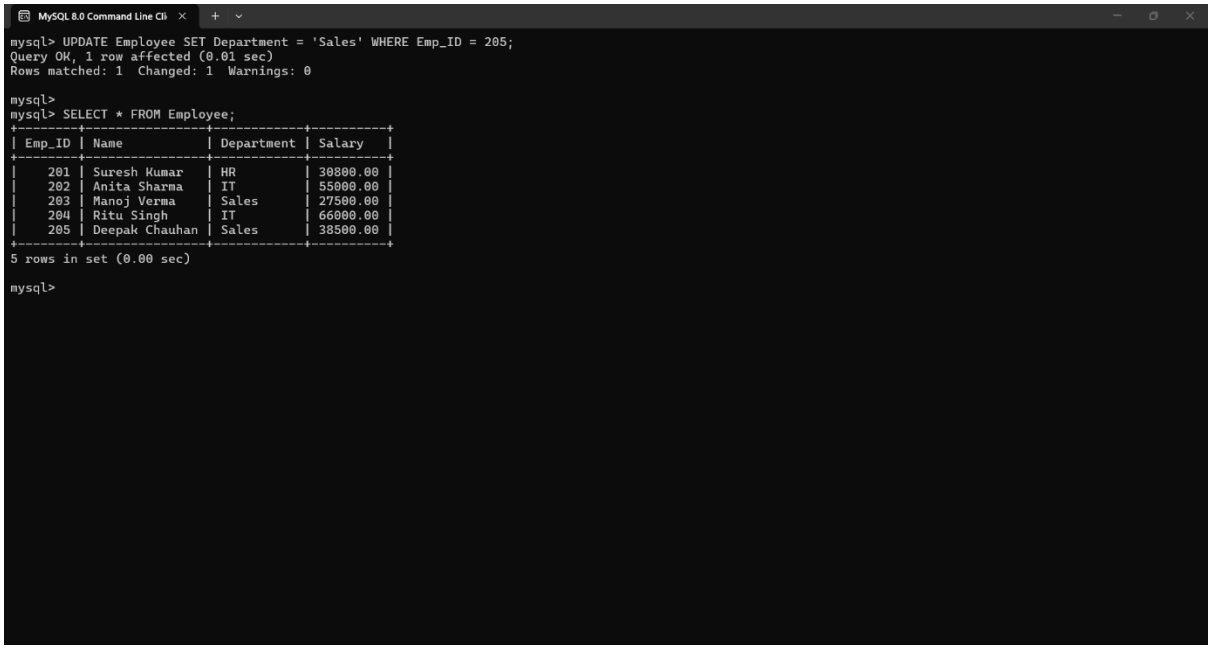
Q15- Change the course of a particular student.

Sol15-



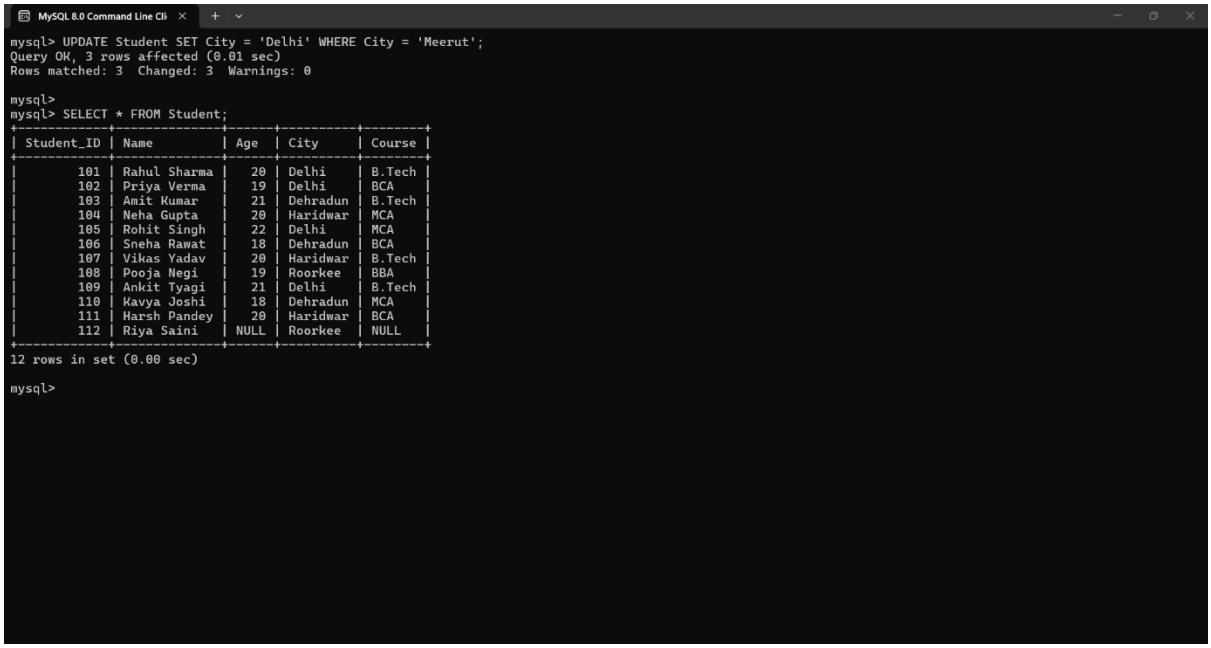
Q16- Update the department of an employee.

Sol16-



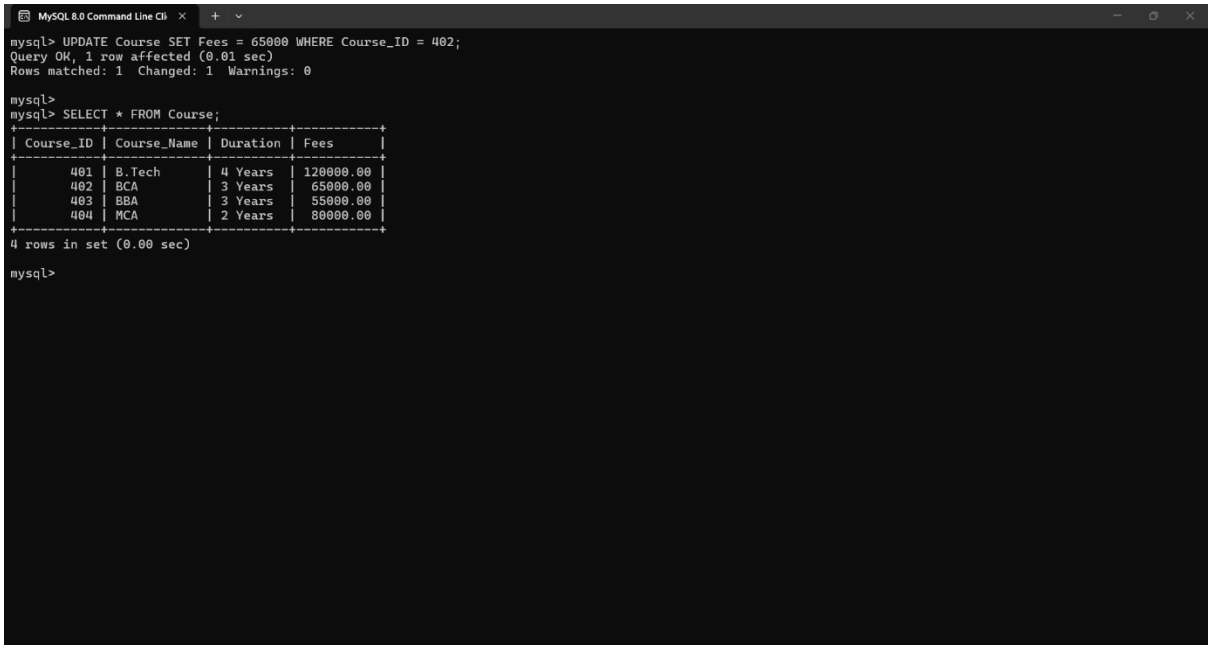
Q17- Update the city of all students from Meerut to Delhi.

Sol17-



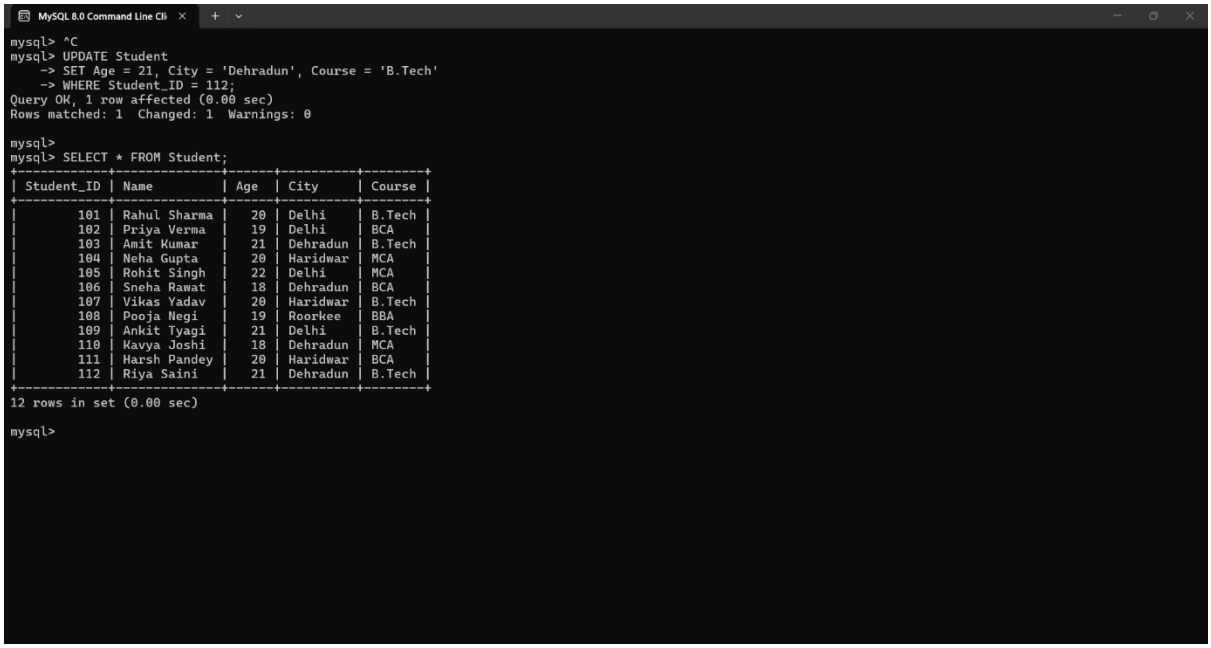
Q18- Update the fees of a particular course.

Sol18-



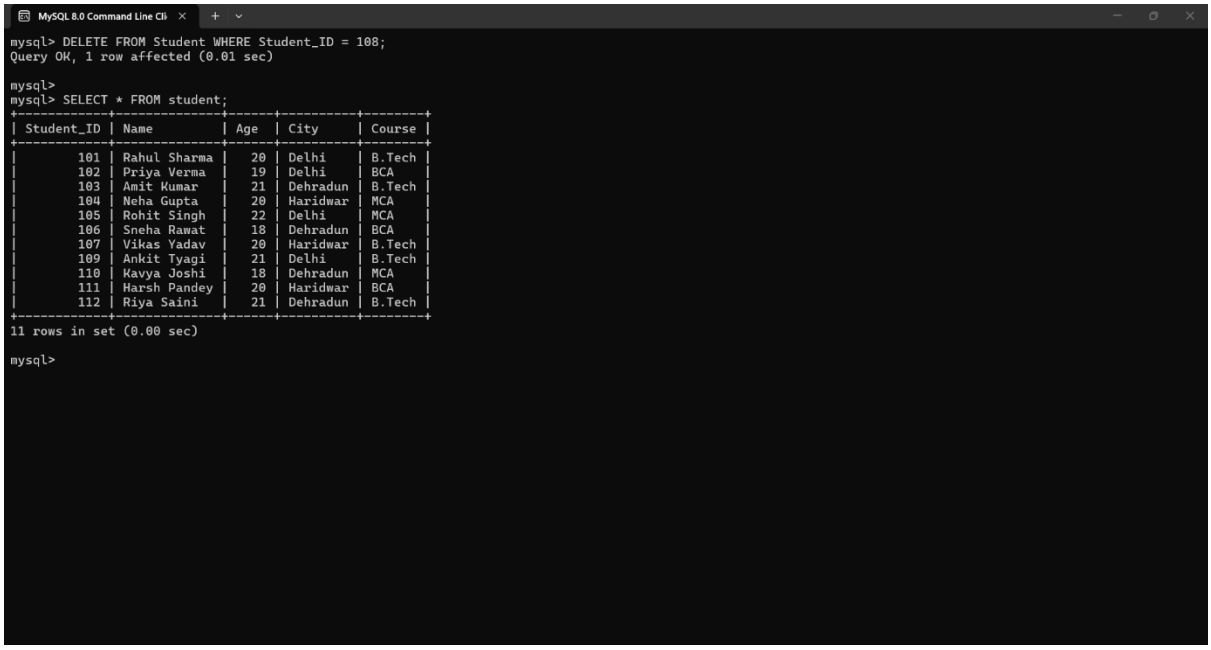
Q19- Update multiple columns of a student simultaneously.

Sol19-



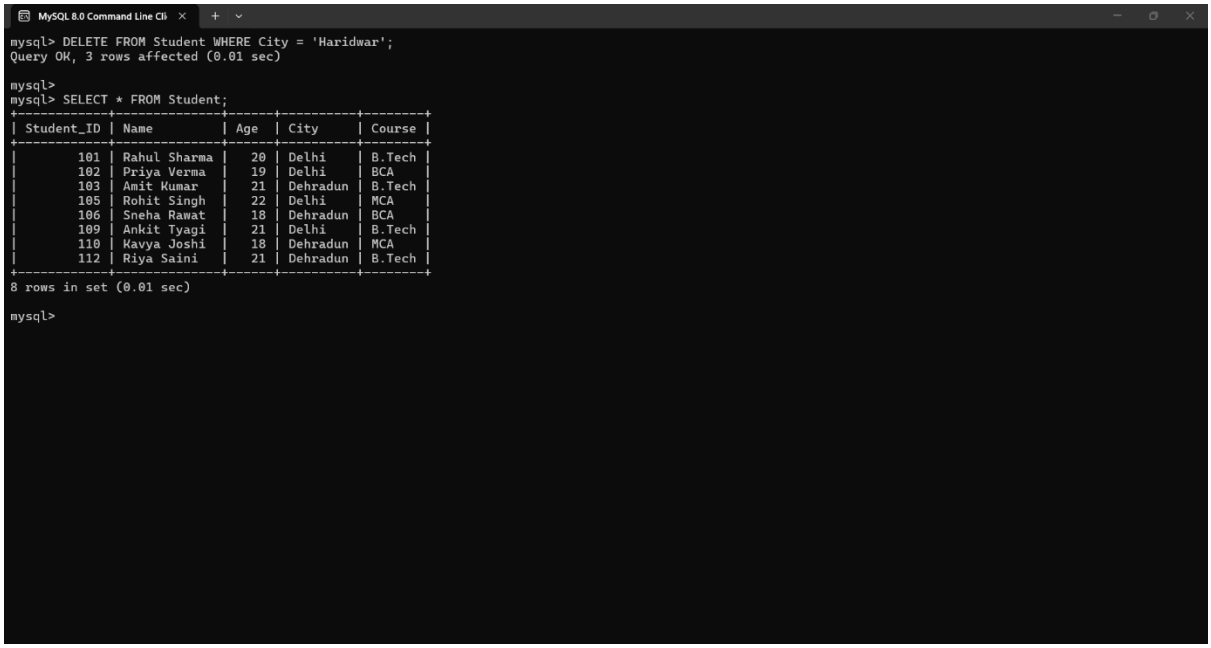
Q20- Delete a particular student record using Student_ID.

Sol20-



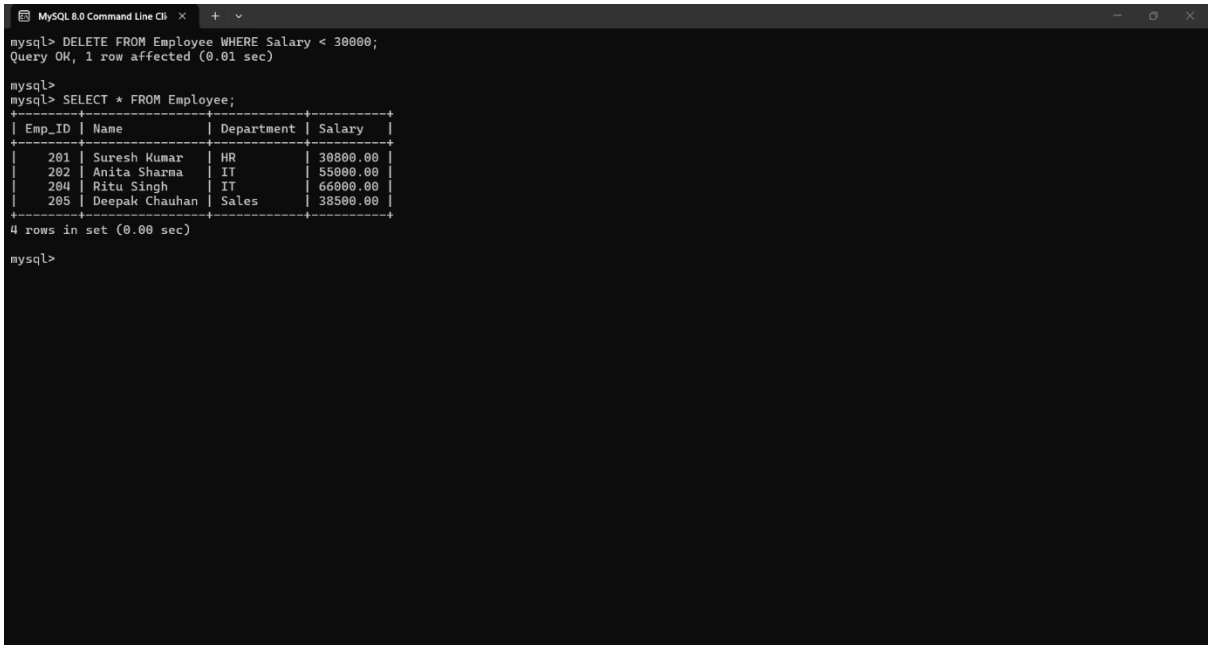
Q21- Delete all students belonging to a particular city.

Sol21-



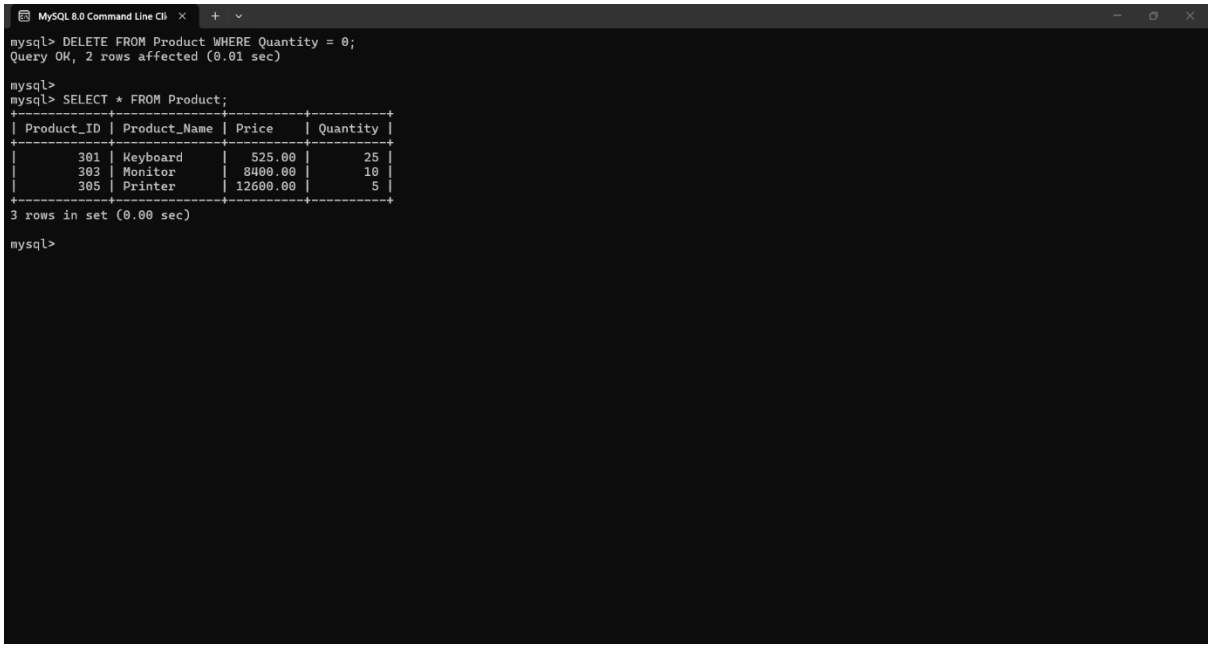
Q22- Delete employees whose salary is below a specified amount.

Sol22-



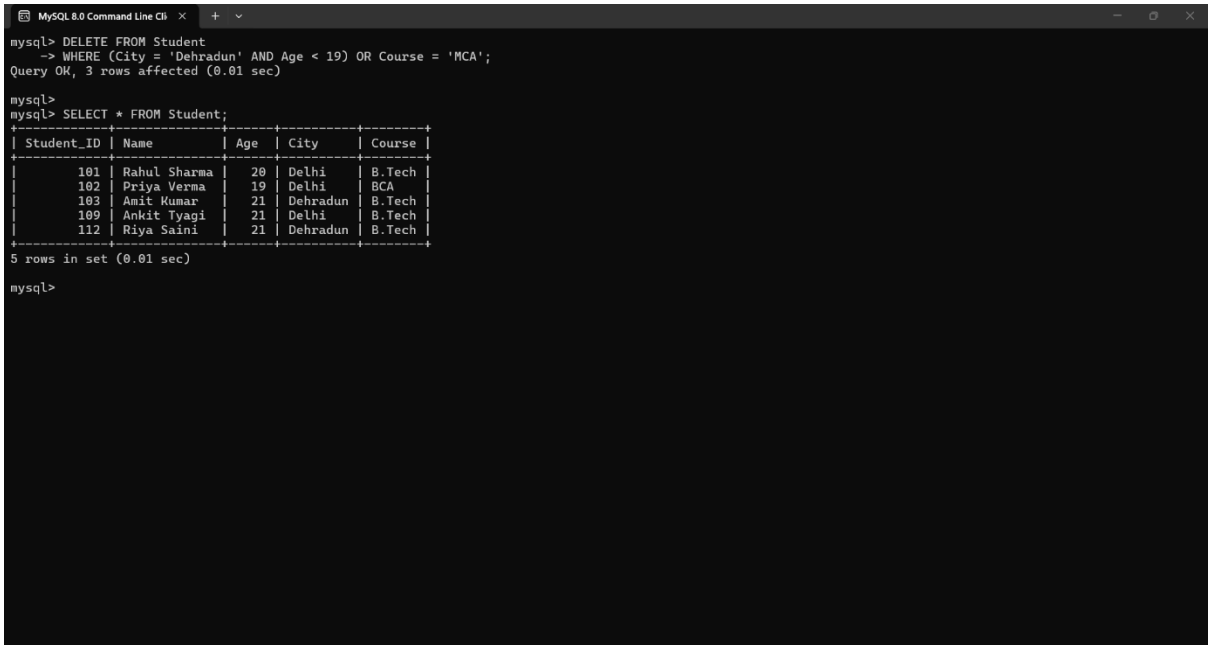
Q23- Delete products whose quantity is zero.

Sol23-



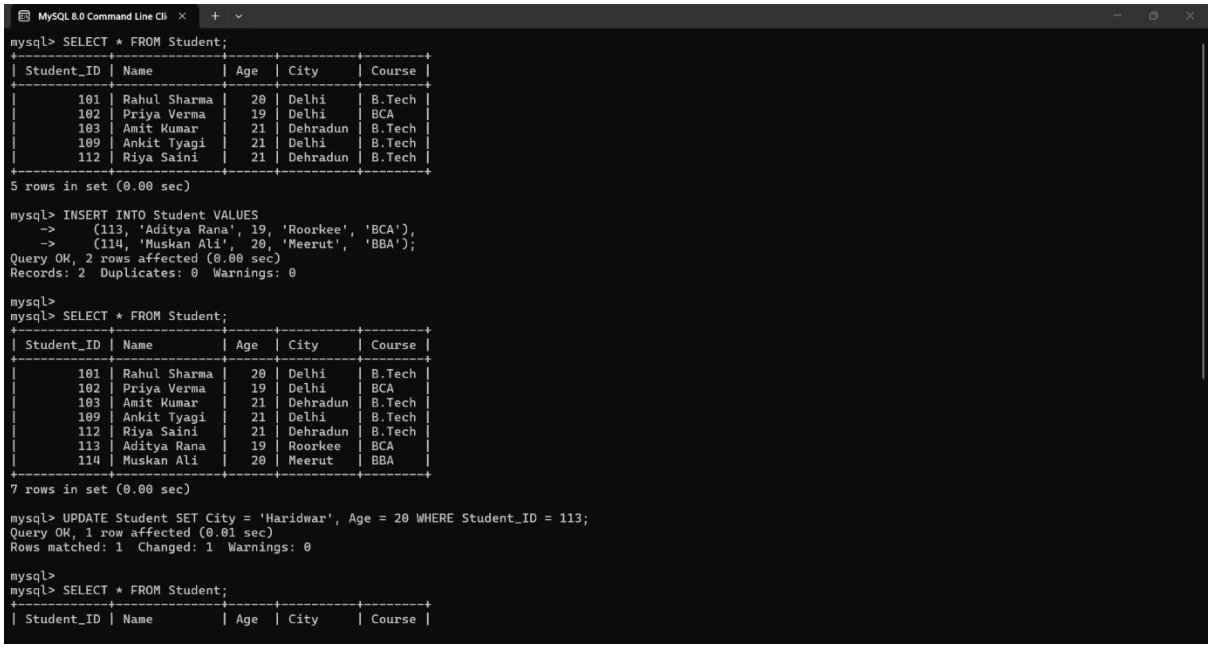
Q24- Delete records satisfying multiple conditions using AND/OR.

Sol24-



Q25- Perform a complete sequence of INSERT, UPDATE and DELETE operations on a Student table and display the table after each operation.

Sol25-



```
MySQL 8.0 Command Line Cli

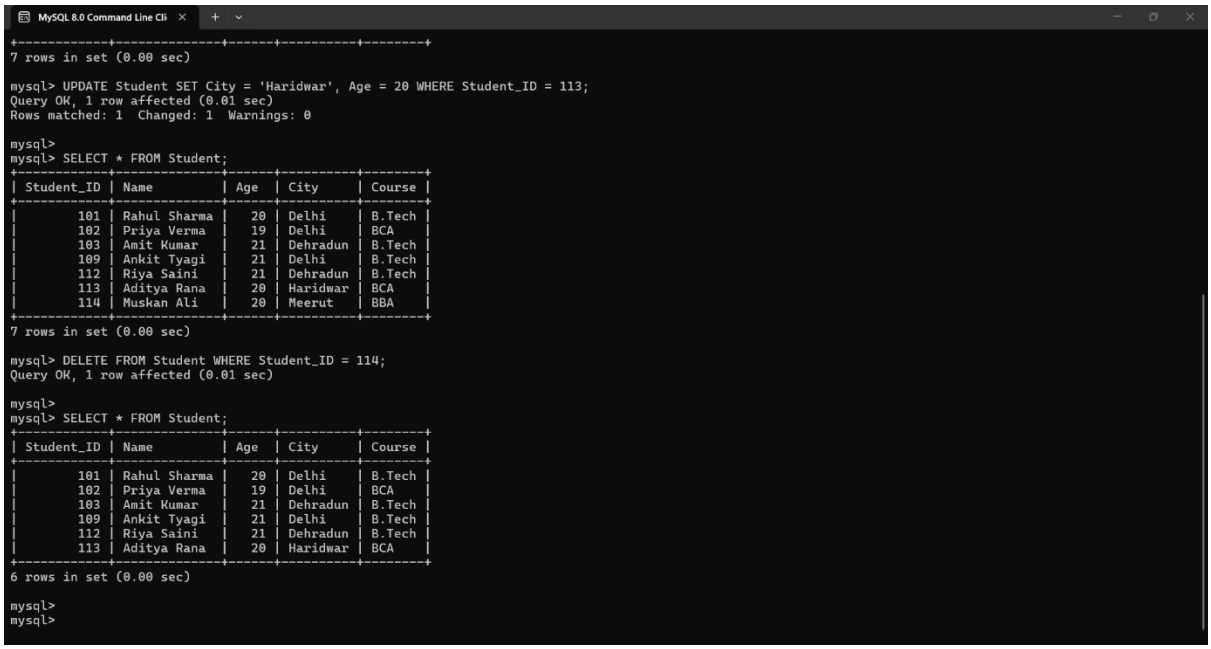
mysql> SELECT * FROM Student;
+------------+--------------+------+----------+--------+
| Student_ID | Name         | Age  | City     | Course |
+------------+--------------+------+----------+--------+
|        101 | Rahul Sharma |   20 | Delhi    | B.Tech |
|        102 | Priya Verma  |   19 | Delhi    | BCA    |
|        103 | Amit Kumar   |   21 | Dehradun | B.Tech |
|        109 | Ankit Tyagi  |   21 | Delhi    | B.Tech |
|        112 | Riya Saini   |   21 | Dehradun | B.Tech |
+------------+--------------+------+----------+--------+
5 rows in set (0.00 sec)

mysql> INSERT INTO Student VALUES
    ->     (113, 'Aditya Rana', 19, 'Roorkee', 'BCA'),
    ->     (114, 'Muskan Ali',  20, 'Meerut',  'BBA');
Query OK, 2 rows affected (0.00 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql>
mysql> SELECT * FROM Student;
+------------+--------------+------+----------+--------+
| Student_ID | Name         | Age  | City     | Course |
+------------+--------------+------+----------+--------+
|        101 | Rahul Sharma |   20 | Delhi    | B.Tech |
|        102 | Priya Verma  |   19 | Delhi    | BCA    |
|        103 | Amit Kumar   |   21 | Dehradun | B.Tech |
|        109 | Ankit Tyagi  |   21 | Delhi    | B.Tech |
|        112 | Riya Saini   |   21 | Dehradun | B.Tech |
|        113 | Aditya Rana  |   19 | Roorkee  | BCA    |
|        114 | Muskan Ali   |   20 | Meerut   | BBA    |
+------------+--------------+------+----------+--------+
7 rows in set (0.00 sec)

mysql> UPDATE Student SET City = 'Haridwar', Age = 20 WHERE Student_ID = 113;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT * FROM Student;
+------------+--------------+------+----------+--------+
| Student_ID | Name         | Age  | City     | Course |
```



```
MySQL 8.0 Command Line Cli

+------------+--------------+------+----------+--------+
7 rows in set (0.00 sec)

mysql> UPDATE Student SET City = 'Haridwar', Age = 20 WHERE Student_ID = 113;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT * FROM Student;
+------------+--------------+------+----------+--------+
| Student_ID | Name         | Age  | City     | Course |
+------------+--------------+------+----------+--------+
|        101 | Rahul Sharma |   20 | Delhi    | B.Tech |
|        102 | Priya Verma  |   19 | Delhi    | BCA    |
|        103 | Amit Kumar   |   21 | Dehradun | B.Tech |
|        109 | Ankit Tyagi  |   21 | Delhi    | B.Tech |
|        112 | Riya Saini   |   21 | Dehradun | B.Tech |
|        113 | Aditya Rana  |   20 | Haridwar | BCA    |
|        114 | Muskan Ali   |   20 | Meerut   | BBA    |
+------------+--------------+------+----------+--------+
7 rows in set (0.00 sec)

mysql> DELETE FROM Student WHERE Student_ID = 114;
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT * FROM Student;
+------------+--------------+------+----------+--------+
| Student_ID | Name         | Age  | City     | Course |
+------------+--------------+------+----------+--------+
|        101 | Rahul Sharma |   20 | Delhi    | B.Tech |
|        102 | Priya Verma  |   19 | Delhi    | BCA    |
|        103 | Amit Kumar   |   21 | Dehradun | B.Tech |
|        109 | Ankit Tyagi  |   21 | Delhi    | B.Tech |
|        112 | Riya Saini   |   21 | Dehradun | B.Tech |
|        113 | Aditya Rana  |   20 | Haridwar | BCA    |
+------------+--------------+------+----------+--------+
6 rows in set (0.00 sec)

mysql>
mysql>
```